

Feature Requirements: Notifications & Reminders

1. Overview & Strategy

Parents often forget to log activities, especially during sleep-deprived stretches. The goal of our notification system is to act as a gentle "nudge," not an alarm.

- V1 Scope: Focus on highly customizable, user-defined scheduled reminders.
- V2 Scope: "Smart" rule-based reminders (e.g., "It's been X hours since last feed").
- Monetization: Custom scheduled reminders are Free. "Smart" predictive reminders and advanced interval alerts are Premium.

2. General Notification Requirements

- Do Not Disturb (DND) Respect: All notifications must instantly mute if the user's phone is in DND mode or if the app detects a "Sleep" timer is currently running.
- Quiet Hours Setting: Users must be able to set a global "Quiet Hours" window (e.g., 10:00 PM to 7:00 AM) where no app notifications are sent, overriding all individual settings.
- Action-Rich Notifications: Android notifications must include quick-action buttons. (e.g., if a "Feed" reminder pops up, the notification itself should have buttons for "Start Nursing" or "Log Bottle" that open directly to that specific screen).
- Snooze/Dismiss: All notifications must allow the user to swipe away (dismiss) or tap "Remind me in 15 mins" (snooze).

3. V1 Requirements: Scheduled Reminders (Free)

Users can manually set alarms to prompt them to log or initiate an activity.

- Trigger Types: Reminders can trigger for: "Time to Feed", "Time to Pump" (if tracking maternal health later), "Time to Sleep/Nap", or general "Check Diaper".
- Recurrence: Users can set reminders as One-Time or Recurring (e.g., "Every 3 hours starting at 8:00 AM").
- Per-Child Targeting: For Premium users with multiple kids, scheduled reminders must be assignable to a specific child (e.g., "Wake up Baby A for feed" vs "Wake up Baby B").

4. V2 Requirements: "Smart" Interval Reminders (Premium)

These reminders trigger dynamically based on the baby's recent history, rather than a strict clock.

- Elapsed Time Alerts:

- "It has been [User-Defined X] hours since the last feeding."
 - "It has been [User-Defined X] hours since the last diaper change."
- Wake Window Alerts:
 - Triggered when a "Nap" or "Night Sleep" timer is stopped.
 - "Baby woke up. Next optimal wake window is approximately [X hours] based on recent averages."
- Maximum Nap Length Alert:
 - "Baby has been sleeping for [X hours]. Recommended to wake to protect the next sleep cycle."
- Logic Constraints: Smart reminders must have a built-in cooldown to prevent spam. (e.g., If a user hits "Snooze" on a feeding reminder, do not send another smart feeding reminder for at least 30 minutes, even if the elapsed time logic would trigger it).

5. User Stories: Notifications

US-NOT-01: Quick Log from Lock Screen

As a nursing mother, I want to tap "Start Nursing" directly from my lock screen notification, so I don't have to unlock my phone, find the app, and navigate to the feeding screen.

Acceptance Criteria: Given I receive a feeding reminder, when I expand the Android notification, I see actionable buttons that deep-link directly into the logging flow.

US-NOT-02: Preventing Nighttime Spam

As a parent whose baby is finally sleeping, I want the app to stop sending notifications, so I am not woken up by a "time to feed" alert when my baby is actually asleep.

Acceptance Criteria: Given I have an active sleep timer running for my baby, when a scheduled feeding reminder is triggered, the notification is suppressed and discarded.

US-NOT-03: Smart Feeding Nudge (Premium)

As a parent who doesn't watch the clock, I want the app to tell me if it's been an unusually long time since a feed, so my baby doesn't get dehydrated.

Acceptance Criteria: Given I am a Premium user, when 4 hours have passed since the last logged feeding, I receive a push notification stating the elapsed time with a quick-log button.